# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** infinitely many solutions (coincident lines) | Compare $a_1/a_2$, $b_1/b_2$, $c_1/c_2$ |
| 2 | **(d)** 5 | Solve by elimination |
| 3 | **(c)** 2 | Solve by elimination |
| 4 | **(d)** 24 | $x = 6$, $y = 4$ |
| 5 | **(d)** 4 | $\frac{1}{2} = \frac{2}{k} \neq \frac{3}{7}$ |
| 6 | **(a)** 2 | Solve, then add |
| 7 | **(d)** no solution (parallel lines) | Compare $a_1/a_2$, $b_1/b_2$, $c_1/c_2$ |
| 8 | **(d)** 4 | Solve by elimination |
| 9 | **(c)** 1 | Solve by elimination |
| 10 | **(b)** 45 | $x = 9$, $y = 5$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Let father $x$, daughter $y$ now (1); $x - 7 = 7(y - 7)$ and $x + 3 = 3(y + 3)$ (1). | 1+1 |
| 12 | Express one variable, substitute: $x = 5$ (1), $y = 3$ (1). | 1+1 |
| 13 | $\frac{1}{2} = \frac{2}{k} \neq \frac{3}{7}$ (1); $k = 4$ (1). | 1+1 |
| 14 | Let father $x$, daughter $y$ now (1); $x - 7 = 6(y - 7)$ and $x + 3 = 3(y + 3)$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\frac{a_1}{a_2} = \frac{5}{7}$, $\frac{b_1}{b_2} = -\frac{2}{3}$, $\frac{c_1}{c_2} = \frac{8}{9}$ (2); ratios $a_1/a_2 \neq b_1/b_2$: lines intersect (1). | 2+1 |
| 16 | Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x = -21$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1). | 2+1+1+1 |
| 17 | Let tens digit $x$, ones digit $y$: $x + y = 9$ (1); $(10y + x) - (10x + y) = -45$ so $y - x = -5$ (1); $x = 7$, $y = 2$, number 72 (1). | 1+1+1 |
| | **OR:** $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$ (2); all ratios equal: lines coincide (1). | 2+1 |
| 18 | Multiply (i) by $-3$ and (ii) by 2 and subtract: $(-13)x = -52$ (2); $x = 4$ (1); put in (i): $y = 1$ (1); check in (ii) (1). | 2+1+1+1 |
| | **OR:** Let tens digit $x$, ones digit $y$: $x + y = 9$ (1); $(10y + x) - (10x + y) = -27$ so $y - x = -3$ (1); $x = 6$, $y = 3$, number 63 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Son $x$, father $8x$ (1); $8x + 9 = 5(x + 9)$ (2); $x = 12$ (1); son 12 years, father 96 years (1). | 1+2+1+1 |
| | **OR:** Fixed $x$, per km $y$: $x + 10y = 170$, $x + 15y = 240$ (2); subtract: $5y = 70$, $y = 14$ (1); $x = 30$ (1); fare for 25 km $= 30 + 14 \times 25 = 380$ (1). | 2+1+1+1 |
| 20 | Son $x$, father $7x$ (1); $7x + 6 = 4(x + 6)$ (2); $x = 6$ (1); son 6 years, father 42 years (1). | 1+2+1+1 |
| | **OR:** Fixed $x$, per km $y$: $x + 8y = 141$, $x + 14y = 213$ (2); subtract: $6y = 72$, $y = 12$ (1); $x = 45$ (1); fare for 30 km $= 45 + 12 \times 30 = 405$ (1). | 2+1+1+1 |

**Mistakes to avoid:**

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.
- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x + y$ again instead of $10y + x$ in digit problems.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 9 | $\frac{1}{3} = \frac{3}{k} \neq \frac{4}{13}$ |
| 2 | **(c)** 7 | Solve, then add |
| 3 | **(a)** exactly one solution (intersecting lines) | Compare $a_1/a_2$, $b_1/b_2$, $c_1/c_2$ |
| 4 | **(a)** 3 | Solve by elimination |
| 5 | **(a)** 2 | Solve by elimination |
| 6 | **(b)** 55 | $x = 11$, $y = 5$ |
| 7 | **(c)** 8 | $\frac{1}{4} = \frac{2}{k} \neq \frac{5}{21}$ |
| 8 | **(b)** 11 | Solve, then add |
| 9 | **(a)** no solution (parallel lines) | Compare $a_1/a_2$, $b_1/b_2$, $c_1/c_2$ |
| 10 | **(a)** 2 | Solve by elimination |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Express one variable, substitute: $x = 4$ (1), $y = 2$ (1). | 1+1 |
| 12 | $\frac{1}{3} = \frac{3}{k} \neq \frac{4}{13}$ (1); $k = 9$ (1). | 1+1 |
| 13 | Let father $x$, daughter $y$ now (1); $x - 7 = 5(y - 7)$ and $x + 3 = 3(y + 3)$ (1). | 1+1 |
| 14 | Express one variable, substitute: $x = 3$ (1), $y = 1$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\frac{a_1}{a_2} = 3$, $\frac{b_1}{b_2} = 3$, $\frac{c_1}{c_2} = \frac{10}{9}$ (2); $a_1/a_2 = b_1/b_2 \neq c_1/c_2$: lines are parallel (1). | 2+1 |
| 16 | Multiply (i) by 2 and (ii) by 3 and subtract: $(-1)x = -3$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1). | 2+1+1+1 |
| 17 | Let tens digit $x$, ones digit $y$: $x + y = 9$ (1); $(10y + x) - (10x + y) = -63$ so $y - x = -7$ (1); $x = 8$, $y = 1$, number 81 (1). | 1+1+1 |
| | **OR:** $\frac{a_1}{a_2} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{1}{3}$, $\frac{c_1}{c_2} = \frac{1}{3}$ (2); all ratios equal: lines coincide (1). | 2+1 |
| 18 | Multiply (i) by 4 and (ii) by 2 and subtract: $(14)x = 28$ (2); $x = 2$ (1); put in (i): $y = 5$ (1); check in (ii) (1). | 2+1+1+1 |
| | **OR:** Let tens digit $x$, ones digit $y$: $x + y = 9$ (1); $(10y + x) - (10x + y) = -9$ so $y - x = -1$ (1); $x = 5$, $y = 4$, number 54 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Let boat $x$ km/h, stream $y$ km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $30p + 45q = 11$, $40p + 27q = 11$ (2); $p = \frac{1}{5}$, $q = \frac{1}{9}$ (1); $x - y = 5$, $x + y = 9$ (1); boat 7 km/h, stream 2 km/h (1). | 2+1+1+1 |
| | **OR:** Length $x$, breadth $y$: $(x - 5)(y + 3) = xy - 30$ gives $3x - 5y = -15$ (1.5); $(x + 3)(y + 2) = xy + 91$ gives $2x + 3y = 85$ (1.5); solving, $x = 20$, $y = 15$ (2). | 1.5+1.5+2 |
| 20 | Let boat $x$ km/h, stream $y$ km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $28p + 30q = 10$, $21p + 50q = \frac{41}{4}$ (2); $p = \frac{1}{4}$, $q = \frac{1}{10}$ (1); $x - y = 4$, $x + y = 10$ (1); boat 7 km/h, stream 3 km/h (1). | 2+1+1+1 |
| | **OR:** Length $x$, breadth $y$: $(x - 4)(y + 2) = xy - 20$ gives $2x - 4y = -12$ (1.5); $(x + 2)(y + 3) = xy + 84$ gives $3x + 2y = 78$ (1.5); solving, $x = 18$, $y = 12$ (2). | 1.5+1.5+2 |

**Mistakes to avoid:**

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.
- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.

- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x + y$ again instead of $10y + x$ in digit problems.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 5 | Solve by elimination |
| 2 | **(c)** 96 | $x = 12$, $y = 8$ |
| 3 | **(c)** 8 | $\frac{1}{2} = \frac{4}{k} \neq \frac{3}{7}$ |
| 4 | **(a)** 11 | Solve, then add |
| 5 | **(b)** infinitely many solutions (coincident lines) | Compare $a_1/a_2$, $b_1/b_2$, $c_1/c_2$ |
| 6 | **(d)** 6 | Solve by elimination |
| 7 | **(d)** 2 | Solve by elimination |
| 8 | **(d)** 128 | $x = 16$, $y = 8$ |
| 9 | **(c)** 15 | $\frac{1}{3} = \frac{5}{k} \neq \frac{2}{7}$ |
| 10 | **(d)** 5 | Solve, then add |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\frac{1}{4} = \frac{2}{k} \neq \frac{5}{21}$ (1); $k = 8$ (1). | 1+1 |
| 12 | Let father $x$, daughter $y$ now (1); $x - 7 = 8(y - 7)$ and $x + 3 = 3(y + 3)$ (1). | 1+1 |
| 13 | Express one variable, substitute: $x = 2$ (1), $y = 5$ (1). | 1+1 |
| 14 | $\frac{1}{2} = \frac{4}{k} \neq \frac{3}{7}$ (1); $k = 8$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{2}{5}$ (2); $a_1/a_2 = b_1/b_2 \neq c_1/c_2$: lines are parallel (1). | 2+1 |
| 16 | Multiply (i) by 2 and (ii) by $-3$ and subtract: $(13)x = 78$ (2); $x = 6$ (1); put in (i): $y = 2$ (1); check in (ii) (1). | 2+1+1+1 |
| 17 | Let tens digit $x$, ones digit $y$: $x + y = 11$ (1); $(10y + x) - (10x + y) = -63$ so $y - x = -7$ (1); $x = 9$, $y = 2$, number 92 (1). | 1+1+1 |
| | **OR:** $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$ (2); all ratios equal: lines coincide (1). | 2+1 |
| 18 | Multiply (i) by $-2$ and (ii) by 4 and subtract: $(-26)x = -104$ (2); $x = 4$ (1); put in (i): $y = 2$ (1); check in (ii) (1). | 2+1+1+1 |
| | **OR:** Let tens digit $x$, ones digit $y$: $x + y = 11$ (1); $(10y + x) - (10x + y) = -27$ so $y - x = -3$ (1); $x = 7$, $y = 4$, number 74 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Let fraction $\frac{x}{y}$: $3(x + 1) = 2(y + 1)$ and $11(x + 3) = 8(y + 3)$ (2); simplify and solve (2); $x = 5$, $y = 8$, fraction $\frac{5}{8}$ (1). | 2+2+1 |
| | **OR:** Let train $u$, bus $v$ km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $240p + 240q = 7$, $300p + 180q = \frac{29}{4}$ (2); $p = \frac{1}{60}$, $q = \frac{1}{80}$ (2); train 60 km/h, bus 80 km/h (1). | 2+2+1 |
| 20 | Let fraction $\frac{x}{y}$: $11(x + 2) = 6(y + 2)$ and $14(x + 5) = 9(y + 5)$ (2); simplify and solve (2); $x = 4$, $y = 9$, fraction $\frac{4}{9}$ (1). | 2+2+1 |
| | **OR:** Let train $u$, bus $v$ km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $160p + 200q = 8$, $200p + 160q = \frac{41}{5}$ (2); $p = \frac{1}{40}$, $q = \frac{1}{50}$ (2); train 40 km/h, bus 50 km/h (1). | 2+2+1 |

**Mistakes to avoid:**

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
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